

S-TSViT: An Energy-Efficient Spiking Transformer for Satellite Image Time Series Analysis

Cao Nguyen Le, University of Information Technology, Vietnam National University, Ho Chi Minh City, Vietnam

Huu Phuoc Dinh, University of Information Technology, Vietnam National University, Ho Chi Minh City, Vietnam

Viet Hang Duong, University of Information Technology, Vietnam National University, Ho Chi Minh City, VietNam * Corresponding author

Abstract — Satellite Image Time Series (SITS) analysis is critical for precision agriculture but demands significant computational resources. In this article, we introduce S-TSViT, a pioneering framework that adapts the efficient factorized temporo-spatial design of transformers into the Spiking Neural Network (SNN) domain. Evaluating on the PASTIS benchmark, the model establishes a new state-of-the-art, achieving a mean Intersection over Union (mIoU) score of 66.9%, while reducing model size by 12% compared to standard Artificial Neural Network (ANN) baselines. Crucially, S-TSViT demonstrates exceptional sparsity with an average firing rate of approximately 6%, resulting in an estimated 85 times reduction in theoretical energy consumption. This work effectively bridges the accuracy gap typically observed between ANNs and SNNs, proving that spike-driven models can deliver high-performance, green AI solutions for Earth observation and resource-constrained deployments.

INTRODUCTION

Research indicates that to ensure global food security by 2050, the agricultural sector must expand by approximately one billion hectares [1]. Consequently, the imperative to monitor plant health and manage agricultural parcels throughout the year has intensified. To address this, Earth Observation (EO) has increasingly integrated with machine learning to automate monitoring processes with minimal human intervention. The urgency of advancing SITS analysis is further underscored by recent regulatory shifts in the European Union (EU). As of January 2023, the Common Agricultural Policy (CAP) mandates that all EU Member States operate the Area Monitoring System (AMS) to

continuously prevent errors and assess policy performance, focusing specifically on agricultural activities [2]. This requirement forces national agencies to assess the quality of agricultural monitoring annually, creating a massive computational demand that current technologies struggle to meet.

Meeting these rigorous policy objectives relies heavily on modern Earth Observation technologies. In recent years, EO capabilities have advanced significantly, driven by the launch of commercial constellations like Planet [3] and open-access missions such as Sentinel-1 and Sentinel-2. These Sentinels provide high-resolution imagery with revisit times as short as five days [4], creating unprecedented opportunities to monitor rapid human interventions, such as harvesting. While Transformers are

architecturally suited for modeling the spatio-temporal dynamics of such data, their deployment on orbital hardware is severely limited by strict power constraints [5]. Therefore, the intersection of efficient Spiking Neural Networks (SNNs) and complex Transformer architectures represents a critical frontier for enabling next-generation, energy-efficient intelligent systems capable of processing both static and sequential visual data.

However, a significant gap remains in applying energy-efficient deep learning to complex sequential data, specifically Satellite Image Time-Series (SITS). SITS data is high-dimensional, containing the rich spatial, spectral, and temporal information required for tasks like crop phenology monitoring. Traditional Transformers (e.g., TSViT) effectively capture these dependencies but are computationally prohibitive for on-board satellite processing. Conversely, current SNN research predominantly focuses on static datasets (like CIFAR/ImageNet) or simple neuromorphic datasets (DVS), leaving the domain of complex, real-world temporal remote sensing largely unexplored. This necessitates a fundamental redesign of the building blocks used for SNN-based time-series analysis.

In this article, the Spiking Temporo-Spatial Vision Transformer (S-TSViT) is proposed, a novel architecture designed for the challenging task of SITS semantic segmentation. This work represents one of the first attempts to adapt Spiking Transformers for complex time-series remote sensing data. The main contributions of this work are summarized as follows:

- Proposing S-TSViT, a pioneering spike-driven architecture that integrates date-aware encoding with a factorized temporo-spatial backbone. This design effectively captures long-range dependencies in irregular time series while maintaining the sparsity of SNNs.
- Establishing a new state-of-the-art for SITS semantic segmentation, achieving the highest mIoU scores on PASTIS (66.9%) benchmarks, outperforming traditional ANN counterparts while reducing parameter count by 12%.
- Achieving significant sparsity with an average firing rate of approximately 6% and theoretically reducing energy consumption by approximately 85 times compared to equivalent ANNs.

BACKGROUND

The dominant paradigm for processing SITS has shifted from Recurrent Neural Networks (RNNs) [6] to Vision

Transformers (ViTs) [7]. Unlike RNNs, which process time steps sequentially, Transformers utilize Self-Attention mechanisms to model global dependencies across the entire time series simultaneously. Specifically, the Temporo-Spatial Vision Transformer (TSViT) [7] has emerged as a leading architecture for crop monitoring. TSViT employs a factorized design: it first processes the temporal dimension to extract phenological evolution (e.g., the growth stages of wheat vs. corn) and then models spatial relationships between neighboring pixels. While effective, this mechanism relies on heavy floating-point matrix multiplications, which scale quadratically with sequence length, creating significant energy bottlenecks for orbital deployment.

Spiking Neural Networks Dynamics To address these energy constraints, Spiking Neural Networks (SNNs) [15] are leveraged, which replace continuous activations with discrete spikes. The fundamental computing unit is the **Leaky Integrate-and-Fire (LIF)** neuron. Its dynamics are governed by three discrete-time phases:

First, the **Membrane Potential Update** accumulates input current $X[t]$ into the membrane voltage $V[t]$ with a decay factor $\beta \in (0,1)$:

$$H[t] = \beta V[t-1] + X[t] \quad (1)$$

Second, the **Spike Generation** mechanism triggers a discrete output spike $S[t]$ if the potential exceeds the firing threshold V_{th} :

$$S[t] = \Theta(H[t] - V_{th}) \quad (2)$$

Finally, a soft reset is employed in the model to preserve the accumulated information for the next time step:

$$V[t] = H[t] - V_{th}S[t] \quad (3)$$

The binary spike $S[t] \in \{0,1\}$ enables the model to possess a sparsity nature, ensuring no downstream task is triggered if no spikes are emitted. For an end-to-end approach, Surrogate Gradient Learning [8] is utilized to mitigate the vanishing gradient challenge in SNNs.

METHODS

Architecture Overview As illustrated in Figure 1, S-TSViT adopts the factorized temporo-spatial design from TSViT [7]. This architectural choice decouples the temporal and spatial dimensions, effectively mitigating the quadratic complexity inherent in standard transformers. The model accepts a 5D input tensor $X \in \mathbb{R}^{B \times T \times C \times H \times W}$. First, the **Temporal Encoder** processes each pixel independently to capture phenological evolution (growth stages) over time. Subsequently, the **Spatial Encoder** models the relationships between

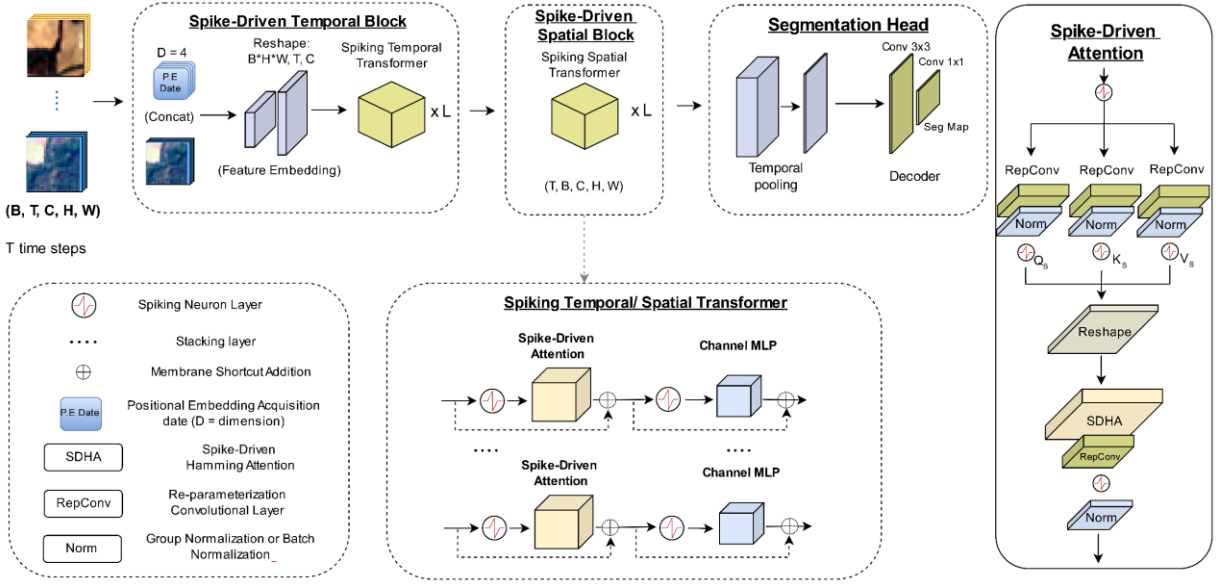


FIGURE 1 . The architecture of the proposed Spiking Temporo-Spatial Vision Transformer (S-TSViT). **(Top)** The pipeline factorizes SITS inputs into a **Spiking Temporal Encoder** to capture phenological evolution and a **Spiking Spatial Encoder** for contextual modeling. **(Bottom-Left)** The annotation of the figure. **(Bottom-Center)** Each transformer block employs a residual pre-activation design where spiking neurons precede the attention layers. **(Right)** The **Spike-Driven Unit** utilizes Re-parameterization Convolutions (RepConv) to generate binary spike maps (Q, K, V), enabling efficient bitwise attention computation.

neighboring pixels to refine contextual boundaries. Finally, the aggregated spike features are passed to the segmentation head for pixel-wise classification.

Date-Aware Input Representation Standard Transformers typically assume fixed temporal intervals between time steps. This assumption is incompatible with SITS, which have irregular revisit cycles of satellites due to orbital dynamics or cloud cover. To address this, a Date-Aware Input Representation is employed. The mechanism is described as:

$$E_{\text{date}} = \text{Embedding}(D; \theta_{\text{date}}) \in \mathbb{R}^{T \times B \times C_{\text{date}}} \quad (4)$$

A learnable lookup table is utilized to project the discrete acquisition day-of-year into a continuous embedding space E_{date} . These embeddings are broadcasted and concatenated with the raw spectral bands along the channel dimension (C):

$$E_{\text{expand}} = \text{Broadcast}(E_{\text{date}}) \in \mathbb{R}^{T \times B \times C_{\text{date}} \times H \times W} \quad (5)$$

$$X_{\text{in}} = \text{Concat}(X, E_{\text{expand}}) \in \mathbb{R}^{T \times B \times (C + C_{\text{date}}) \times H \times W} \quad (6)$$

A convolutional layer then projects this combined input into a unified feature space of dimension d_{model} , ensuring

the network is conditioned on the seasonality of the vegetation from the very first layer.

Cloud Masking Strategy To mitigate the impact of atmospheric noise common in massive region datasets like T31TFM-1618, we implemented a thresholding mechanism. Pixel vectors where all visible channels (R, G, B) exceed a threshold τ are masked to the zero vector, preventing noise-corrupted pixels from disrupting the SNN's temporal integration. The operation is defined as:

$$X_{t,h,w} = \mathbf{0} \quad \text{if } (X_{t,R} > \tau) \wedge (X_{t,G} > \tau) \wedge (X_{t,B} > \tau) \quad (7)$$

where t, h, w denote Time, Height, and Width respectively, τ is the threshold, R, G, B stand for RGB channels.

Spiking Temporal Encoder The temporal encoder is composed of a stack of L identical spiking transformer blocks that process the reshaped input data $X \in \mathbb{R}^{B \cdot H \cdot W \times T \times D}$. To overcome the vanishing gradient problem common in deep SNNs, S-TSViT adapts the transformer module from SpikeVideoFormer (SVF) [9], utilizing a pre-activation residual structure. Formally, the signal propagation is defined as:

Model	#params. (M)	PASTIS	T31TFM-1618
BiCGRU	4.5	80.5 / 56.2	88.6 / 57.7
FPN-CLSTM	1.2	81.9 / 59.5	88.4 / 57.8
UNET3D	6.2	82.3 / 60.4	88.4 / 57.6
UNET3Df	7.2	82.1 / 60.2	88.6 / 57.7
UNET2D-CLSTM	2.3	82.7 / 60.7	89.0 / 58.8
U-TAE	1.1	82.9 / 62.4 (83.2 / 63.1)	88.9 / 58.5
TSViT	1.7	83.4 / 65.1 (83.4 / 65.4)	90.3 / 63.1
S-TSViT (Ours)	1.5	81.7 / 67.3 (81.8 / 66.9)	84.4 / 58.0

TABLE 1. Comparative benchmarking results on the PASTIS and T31TFM-1618 datasets. For the PASTIS dataset, values represent Fold-1 performance, with the 5-fold cross-validation average provided in parentheses. **Bold** values indicate the highest performance in each column.

$$X'_l = X_l + \text{SDHA}(\text{LIF}(X_l)) \quad (8)$$

$$X_{l+1} = X'_l + \text{MLP}(\text{LIF}(X'_l)) \quad (9)$$

where $\text{LIF}(\cdot)$ denotes the Multi-Step Leaky Integrate-and-Fire neuron, SDHA denotes the Spike-Driven *Hamming* Attention module. By placing the spiking neuron before the operations, the main residual path (X_l) is ensured to carry floating-point membrane potentials, preserving information flow, while the processing branches operate on sparse binary spikes.

Spike-Driven Hamming Attention (SDHA) Standard self-attention entails a quadratic complexity $O(T^2)$ and high memory cost for long sequences. To address this, S-TSViT leverages the Linear Hamming attention mechanism introduced in SpikeVideoFormer (SVF) [9], which reformulates the attention operation to achieve linear complexity $O(T)$. The attention output is computed as:

$$\text{SDHA} = \text{LIF}_{2D} \left(\underbrace{(2Q_s - 1)[(2K_s - 1)^T V_s]}_{O(ND^2)} \right) \quad (10)$$

where LIF_{2D} denotes a spiking neuron with a threshold scaled by 2D. Binary spikes are converted into bipolar domain $\{-1, 1\}$. By computing the term $((2K - 1)^T \cdot V)$ first, the operation avoids computing a massive $T \times T$ attention map. This allows the model to efficiently process long temporal sequences on edge devices.

Spiking Spatial Encoder Before spatial processing, the tensor is reshaped to $X \in R^{(T \times B \times C \times H \times W)}$. This effectively swaps the temporal and spatial dimensions, enabling the model to capture the spatial context between neighboring pixels. The Spiking Spatial Encoder also comprises a stack of L spiking transformer blocks identical to the temporal stage, thereby minimizing the complexity of the S-TSViT architecture.

$$X_{out} = \underbrace{\text{SpatialBlock}(\dots \text{SpatialBlock}(X_{in}))}_{L \text{ times}} \quad (11)$$

Prediction Head and Optimization To generate the final segmentation map, the model performs Global Temporal Pooling to create a static 2D feature map. Subsequently, the data is passed to a lightweight Convolutional

Decoding Head. The averaging of spikes and the CNN layers is formalized as:

$$F_{\text{global}} = \frac{1}{T} \sum_{t=1}^T S_{\text{enc}}[t], \quad X \in \mathbb{R}^{B \times C \times H \times W} \quad (12)$$

$$F_{\text{feat}} = \text{ReLU} \left(\text{Norm} \left(\text{Conv}_{3 \times 3} (F_{\text{global}}) \right) \right) \quad (13)$$

$$Y_{\text{pred}} = \text{Conv}_{1 \times 1} (F_{\text{feat}}) \in \mathbb{R}^{B \times N_{\text{classes}} \times H \times W} \quad (14)$$

where $S_{\text{enc}}[t]$ denotes the output spike tensor from the final encoder, F_{global} represents the global pooling feature map, Y_{pred} denotes the final segmentation map logit.

Given the extreme class imbalance inherent in agricultural datasets, standard Cross-Entropy often biases the model toward majority classes. To address this, we utilized Focal Loss [10] for S-TSViT. A modulating factor $(1 - p_t)^\gamma$ is added into the loss function to down-weight easy examples and force the model to focus on the hard, misclassified pixels typical of rare vegetation classes. The loss function is described as:

$$\mathcal{L}_{\text{Focal}} = -\alpha_t (1 - p_t)^\gamma \log(p_t) \quad (15)$$

where p_t denotes the predicted probability, α_t represents the weighting factor, γ denotes the focusing parameter.

EXPERIMENTS AND ANALYSIS

The proposed S-TSViT is evaluated on two publicly available satellite benchmarks: PASTIS [11] (24x24) and T31TFM-1618. The primary benchmark, PASTIS, consists of 2,433 patches, each representing a 1.28×1.28 km area in France. The dataset comprises 18 crop classes and typically has 38-61 acquisitions per year. This irregular temporal sampling serves as a validation ground for the model. To further assess robustness on large-scale landscapes, the T31TFM-1618 dataset is utilized. Unlike the curated PASTIS patches, this dataset covers a massive contiguous area (110×110 km) in Southwestern France spanning 2016-2018. It serves as a stress test for handling frequent cloud cover and atmospheric artifacts across 19 diverse crop classes.

Implementation Details S-TSViT is implemented using the Pytorch [12] and SpikingJelly [13] framework on an NVIDIA RTX 3090 GPU. The model is trained for 50 epochs on PASTIS and 90 epochs for T31TFM-1618, using the AdamW optimizer with an initial learning rate of $1e^{-3}$ and cosine annealing. To prevent overfitting on the limited dataset, standard geometric augmentations are

applied, including random vertical/horizontal flips, noise injection, and 90-degree rotations. Input spectral bands are pre-processed using Z-score standardization to stabilize the spiking thresholds.

Due to the memory constraints of processing high-dimensional satellite time-series, restricted batch sizes are utilized, specifically setting $B = 3$ for the PASTIS and $B = 5$ for the T31TFM-1618. Consequently, Group Normalization (GN) is employed instead of Batch Normalization, which is unstable at small batch sizes. To mitigate the "Zero-Shift" phenomenon—where zero-padding biases the normalization statistics and generates spurious "ghost spikes"—an analysis of padding zero rates is conducted. Based on this experiment, the sequence length is set to $T = 49$ for PASTIS, which is the optimal length for minimizing the padding error problem.

Comparative Results As presented in Table 1, S-TSViT establishes a new state-of-the-art on the PASTIS benchmark, achieving a Mean Intersection over Union (mIoU) of 66.9% averaged across five folds. This performance surpasses the leading ANN-based competitor, TSViT (65.4%), by a margin of 1.5 percentage points.

Notably, while the Overall Accuracy (OA) of 81.8% is slightly lower than most of the baselines, this reflects a deliberate optimization strategy. The use of Focal Loss mitigates the bias toward majority classes, compelling the model to prioritize the learning of minority class signatures rather than overfitting to the background. This result is particularly significant given the inherent constraints of neuromorphic computing; S-TSViT outperforms its full-precision ANN counterpart while exhibiting greater parameter efficiency (1.5M vs. 1.7M parameters).

Energy Efficiency To quantify the hardware advantages of the spiking architecture, the theoretical energy consumption is analyzed using standard 45nm CMOS energy benchmarks [14]. The energy metrics are defined as follows:

$$E_{\text{ANN}} = \text{FLOPs} \times E_{\text{MAC}} \quad (16)$$

$$E_{\text{SNN}} = \text{FLOPs}(T) \times f_r \times E_{\text{AC}} \quad (17)$$

$$\text{Ratio} = \frac{E_{\text{MAC}}}{f_r \times E_{\text{AC}}} \quad (18)$$

where FLOPs denotes the total floating-point operations, FLOPs(T) represents the total operations accumulated

Metric	Value
Average Firing Rate (fr)	5.97%
ANN Static FLOPs (per sample)	37.93 G
ANN Energy (EMAC)	174.46 mJ
SNN Energy (EAC)	2.04 mJ
Efficiency Gain	$85.5 \times$

TABLE 2. Energy Efficiency Profile (1.49M Model, average 5 fold PASTIS).

over T time steps, $E_{MAC} = 4.6 pJ$ denotes the energy consumption of one multiply-accumulate operation, $E_{AC} = 0.9 pJ$ denotes the energy consumption of one accumulate operation, and f_r denotes the firing rate of the model.

With the extremely sparse average firing rate of 5.97% in Table 2, S-TSViT can reduce energy consumption by approximately 85.67 times compared to an equivalent ANN, validating the suitability of the model in resource-constrained environments, such as onboard satellite processing.

ABLATION STUDIES

To validate the individual contributions of our proposed architecture, a series of experiments are conducted analyzing the impact of each major component. Unless otherwise specified, all of the ablation studies in this section are trained on a representative 10% subset of the data and evaluated on a separate 5% validation set.

Impact of Hamming Attention A core hypothesis of this work is that Hamming operations [9] can replace expensive dot products without compromising accuracy. To verify this, a comparative experiment is conducted using Hamming and dot-product attention on a representative 30% subset of the training data and evaluated on a separate 5% validation set.

As illustrated in the supplementary materials, the Hamming mechanism yields superior performance in the majority of parcel classes. We observe particularly significant IoU improvements in minority classes such as Sorghum (+0.192) and Winter Triticale (+0.115). This confirms that converting binary spikes into bipolar preserves critical spike information often lost in standard binary encoding, thereby enhancing the model's segmentation power.

Furthermore, our energy analysis demonstrates that SDHA promotes greater temporal sparsity. The Hamming-based model achieves a lower average firing rate (6.27%) compared to the dot-product variant (6.83%). Consequently, SDHA achieves a higher theoretical energy efficiency gain of 81.56 times (vs. 74.84 times for dot-product), proving that Hamming distance is both more accurate and more hardware-friendly for SITS analysis.

Impact of Focal Loss SITS datasets inherently suffer from extreme class imbalance, reflecting real-world agricultural landscapes where dominant vegetation covers vast areas compared to rare crops. To mitigate this, Focal Loss [10] is integrated as a replacement for standard Cross-Entropy (CE).

As illustrated in supplementary materials, the Focal Loss objective yields superior segmentation performance across nearly all vegetation classes. Substantial improvements are recorded in specific crop types, such as Meadow (+0.308), Sunflower (+0.246), and Winter Rapeseed (+0.220). While performance on the majority "Background" class dips slightly (-0.039), this is a desirable trade-off: the model shifts its learning capacity away from easy, dominant pixels to focus on distinguishing difficult, minority crops, thereby significantly improving the macro-averaged semantic quality.

Impact of Temporal Encoder The Spiking Temporal Encoder is critical for capturing the unique phenological evolution of different crop types. To quantify its effectiveness, an ablation study is performed by removing this module.

As expected, excluding the temporal block results in a significant performance degradation, with mIoU dropping from 0.53 to 0.48 (a loss of 5 percentage points). This decline confirms that static spatial features alone are insufficient for distinguishing crops with similar spectral signatures; the model relies heavily on analyzing the temporal growth cycle to resolve ambiguities between parcel classes.

Impact of Spatial Encoder The Spiking Spatial Encoder is designed to capture local spatial context and field boundaries, adopting the factorized temporo-spatial architecture of TSViT [7]. To evaluate its specific contribution, the model's performance is assessed without the spatial block.

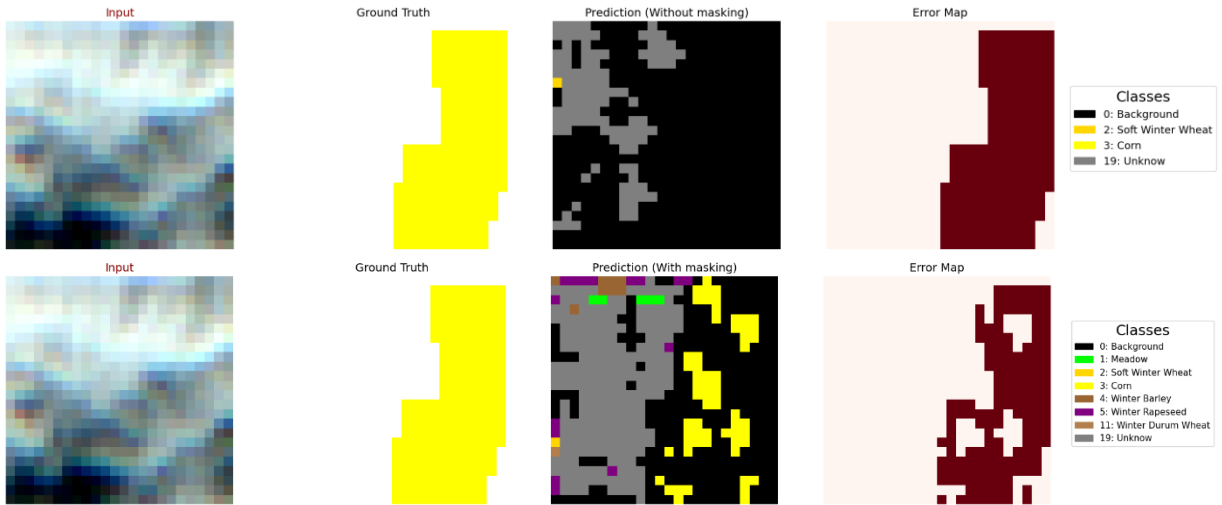


FIGURE 2 . Qualitative comparison of model predictions under severe cloud cover. (Top) Prediction from the baseline S-TSViT (without masking) shows high error rates. (Bottom) Prediction from the S-TSViT with Cloud Masking ($\tau = 3000$) demonstrates improved structural recovery and reduced error, as the model effectively utilizes temporal context to ignore the occluded data.



FIGURE 3 . Visualization of spatial failure in a complex agricultural parcel. The model fails to resolve the minority class Spring Barley (Purple), overwriting it with predictions of the dominant neighboring classes.

Interestingly, removing the spatial encoder results in a marginal mIoU decrease of only 0.19% (from 52.96% to 52.77%). This result suggests that for SITS, the temporal evolution of spectral bands is the primary driver of classification performance. However, under the reduced data regime (10% subset), complex spatial dependencies may not be fully captured, as effective generalization typically requires a larger diversity of training examples.

Impact of Cloud Masking The impact of this strategy is visually demonstrated in Figure 2. The baseline model (Top Row), operating without masking, struggles with noise-corrupted inputs, often failing to extract meaningful spatial structures or hallucinating incorrect classes in cloudy regions. In contrast, the model trained with the cloud masking mechanism (Bottom Row) successfully recovers the underlying semantic segmentation map. As shown in the Error Maps, the masked model reduces false negatives in the crop fields (Yellow regions), proving that removing noisy timestamps allows the Temporal

Attention mechanism to focus more effectively on clear, informative frames from the rest of the time-series.

Quantitatively, this mechanism improves the mIoU on the T31TFM-1618 dataset from 55% (baseline) to 58%. Furthermore, we conducted an ablation study to assess feature robustness. When the masking mechanism is disabled during inference for the model trained with masking, the performance drops only slightly to 57.5%. This result is significant: it indicates that the cloud masking strategy acts as a powerful regularizer during training. It forces the model to learn more generalizable and robust temporal features, preventing it from overfitting to atmospheric noise, even when the safety filter is removed at test time.

Error Analysis Despite the robust global performance of S-TSViT, a qualitative analysis reveals specific failure modes in highly heterogeneous landscapes. As illustrated in Figure 3, the model performance degrades significantly in complex regions where multiple small parcel types coexist in a dense arrangement.

The primary source of error is "Class Dominance". In this example, the model fails to resolve the minority class Spring Barley (Purple), mistakenly overwriting it with predictions of the dominant neighboring classes. This suggests that while Focal Loss successfully addresses global dataset imbalance, the model still suffers from local spatial bias in ambiguous boundary zones. The spatiotemporal factorization, while efficient, appears to over-smooth high-frequency spatial details, causing the "erosion" of minority parcels into the surrounding majority background. This limitation indicates that further refinement in boundary-aware loss functions or higher-resolution spatial encoders is necessary to fully resolve such complex agricultural zone.

Additionally, the model faces challenges regarding cloud cover resilience. Although the proposed cloud masking mechanism improves performance to 0.58 mIoU, it notably lags behind the full-precision TSViT baseline (0.63 mIoU). This gap indicates that while the current thresholding strategy provides necessary noise reduction, it is insufficient for complete restoration, suggesting the need for more advanced cloud removal mechanisms in the future.

CONCLUSION

In this article, the Spiking Temporo-Spatial Vision Transformer (S-TSViT) is proposed as a pioneering spike-based architecture designed for efficient Satellite Image Time Series (SITS) analysis. By integrating a novel date-aware encoding with a factorized temporo-spatial backbone and a spike-driven self-attention mechanism, S-TSViT establishes a new state-of-the-art for semantic segmentation on the PASTIS benchmark. Crucially, the model outperforms traditional ANN competitors while achieving a 12% reduction in parameters and a theoretical energy efficiency gain of up to 85 times. These results bridge the gap between high-performance deep learning and the strict power constraints of edge computing, validating the feasibility of deploying complex neuromorphic architectures directly on orbital devices. Future work will focus on three key areas: resolving the local "majority class dominance" in heterogeneous landscapes, improving cloud cover robustness, and further optimizing the training latency for rapid deployment.

ACKNOWLEDGEMENTS

This research was supported by The VNUHCM - University of Information Technology's Scientific Research Support Fund.

REFERENCES

1. P. Gong et al., "Finer resolution observation and monitoring of global land cover: First mapping results with Landsat TM and ETM+ data," *Int. J. Remote Sens.*, vol. 34, no. 7, pp. 2607–2654, 2013.
2. European Commission, "CAP paying agencies," 2023. [Online]. available: https://agriculture.ec.europa.eu/common-agricultural-policy/financing-cap/cap-paying-agencies_en. [Accessed: Jan. 1, 2026].
3. C. Boshuizen, J. Mason, P. Klupar, and S. Spanhake, "Results from the Planet Labs Flock constellation," in *Proc. AIAA/USU Conf. Small Satellites*, Logan, UT, USA, 2014.
4. M. Drusch et al., "Sentinel-2: ESA's optical high-resolution mission for GMES operational services," *Remote Sens. Environ.*, vol. 120, pp. 25–36, 2012.
5. Y. Pang, Y. Zhang, Q. Kong, Y. Wang, B. Chen, and X. Cao, "SOCDet: A lightweight and accurate oriented object detection network for satellite on-orbit computing," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–15, 2023.
6. D. Ienco, R. Gaetano, C. Dupaquier, and P. Maurel, "Land cover classification via multitemporal spatial data by deep recurrent neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1685–1689, Oct. 2017.
7. M. Tarasiou, E. Chavez, and S. Zafeiriou, "ViTs for SITS: Vision Transformers for Satellite Image Time Series," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Vancouver, BC, Canada, 2023, pp. 10418–10428.
8. O. Neftci, H. Mostafa, and F. Zenke, "Surrogate gradient learning in spiking neural networks: Bringing the power of gradient-based optimization to spiking neural networks," *IEEE Signal Process. Mag.*, vol. 36, no. 6, pp. 51–63, Nov. 2019.
9. S. Zou et al., "SpikeVideoFormer: An efficient spike-driven video transformer with Hamming attention and $\mathcal{O}(T)$ complexity," *arXiv preprint arXiv:2505.10352*, 2025.
10. T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Venice, Italy, 2017, pp. 2980–2988.

11. V. Sainte Fare Garnot and L. Landrieu, "Panoptic segmentation of satellite image time series with convolutional temporal attention networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Montreal, QC, Canada, 2021, pp. 4852–4861.
12. A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, Vancouver, BC, Canada, 2019, pp. 8024–8035.
13. W. Fang et al., "SpikingJelly: An open-source machine learning infrastructure for SNN," *Sci. Adv.*, vol. 9, no. 40, Art. no. eadi1480, Oct. 2023.
14. M. Horowitz, "Computing's energy problem (and what we can do about it)," in *Proc. IEEE Int. Solid-State Circuits Conf. (ISSCC)*, San Francisco, CA, USA, 2014, pp. 10–14.
15. W. Maass, "Networks of spiking neurons: The third generation of neural network models," *Neural Netw.*, vol. 10, no. 9, pp. 1659–1671, Dec. 1997.

Cao Nguyen Le is currently working toward the B.S. degree in the Faculty of Computer Science in Information Technology University-VNU HCM, HoChiMinh, Viet Nam. His research interests include image processing, computer vision, and deep learning. Contact him at 22520970@gm.uit.edu.vn.

Huu Phuoc Dinh is currently working toward the B.S. degree in the Faculty of Computer Science in Information Technology University-VNU HCM, HoChiMinh, Viet Nam. His research interests include image/video processing, computer vision, and deep learning. Contact him at 22521150@gm.uit.edu.vn.

Viet-Hang Duong received the B.S., M.S. degree in applied analysis mathematics from the Can Tho University, Viet Nam, in 1999, 2003, and Ph.D. degree in Computer Science and Information Engineering in the National Central University, Taoyuan, Taiwan, in 2017. She is currently working as lecture in the Faculty of Computer Science in Information Technology University-VNU HCM, HoChiMinh, Viet Nam. Her research interests include computer vision, machine learning, and natural language processing. Contact her at hangdv@uit.edu.vn.